Development of an Intelligent Assistant for Data Preparation Automation in Urban Transportation Management Systems

Thesis for the Award of the Degree/Diploma of
MASTER
By

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Abstract

This dissertation presents comprehensive research on the development of an intelligent assistant for data preparation automation in urban transportation management systems. The research addresses the critical challenge of automating data preparation workflows that currently consume approximately 80% of data scientists' time according to industry surveys.

The proposed framework integrates multi-dimensional data quality assessment, automated feature engineering, ensemble anomaly detection, and traffic event classification within a unified system. The methodology encompasses sophisticated mathematical formulations for data quality metrics, spatiotemporal feature engineering, ensemble-based anomaly detection algorithms, and advanced machine learning classification models.

Experimental evaluation using semi-synthetic traffic data from Astana, Kazakhstan demonstrates the effectiveness of the proposed approach. The results show a data quality index of 0.957, classification accuracy of 96.62%, and effective anomaly detection with an ensemble method identifying 3.92% of data as anomalies.

The feature engineering process generates 16 derived features from 9 original attributes, capturing essential spatiotemporal patterns. Random Forest classification achieved the best performance with 96.62% accuracy, with key predictive features being Congestion Probability (30.42%), Traffic Density (26.67%), and Flow Rate (8.41%).

The findings have significant implications for urban transportation management, suggesting that automated data preparation can substantially improve data quality, reduce manual effort, and enable more sophisticated traffic analytics in smart city environments.

Keywords: Data Preparation, Intelligent Transportation Systems, Machine Learning, Anomaly Detection, Smart Cities, Traffic Management

Dedication

Dedicated to all researchers and practitioners working towards making our cities smarter and more livable through advancements in intelligent transportation systems.

Declaration

I hereby declare that this dissertation is my own work and has been conducted independently. The research has not been submitted previously for examination, nor has been accepted for publication.

I certify that this dissertation represents original work and has not incorporated material from other sources except where explicitly acknowledged and referenced in accordance with proper academic standards.

The work in this dissertation was the result of my own investigation and except where specifically acknowledged by reference.

I have read and understood the university's policy on plagiarism and am aware of the consequences of this action.

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Chapter 1

Introduction

1.1 Background and Motivation

The rapid urbanization occurring globally has placed unprecedented demands on urban transportation infrastructure, necessitating the development of sophisticated Intelligent Transportation Systems (ITS) that can efficiently manage traffic flow, detect incidents, and optimize mobility services. As urban populations continue to expand, transportation authorities face mounting challenges in processing vast quantities of heterogeneous data generated by an increasingly diverse array of sensors, including inductive loop detectors, video surveillance systems, GPS-enabled vehicles, connected infrastructure, and emerging Internet of Things (IoT) devices deployed throughout modern smart cities.

The evolution from traditional traffic management approaches to intelligent, data-driven systems represents a fundamental paradigm shift in how cities approach transportation challenges. Modern ITS deployments generate enormous volumes of multi-modal, multi-source data streams that require sophisticated processing pipelines capable of handling real-time ingestion, transformation, quality assessment, and analytical processing before the data can be effectively utilized for decision-making purposes.

Central to this transformation is the critical role of data preparation—the foundational process that determines the quality, reliability, and ultimately the utility of all downstream analytical and machine learning applications. Research consistently demonstrates that data scientists and transportation analysts spend approximately 80% of their time on data preparation tasks, including cleaning, transformation, integration, and quality assessment, rather than on value-generating analytical activities.

1.2 Problem Statement

Despite significant advances in machine learning algorithms and computational infrastructure for transportation analytics, the data preparation process remains predominantly manual, time-consuming, and error-prone. Transportation datasets are characterized by several inherent challenges that complicate automated processing: high dimensionality arising from multiple sensor modalities, temporal dependencies spanning various granularities from seconds to seasonal patterns, spatial correlations across network topologies, missing values due to sensor malfunctions or communication failures, and heterogeneous formats requiring extensive normalization and standardization.

The absence of systematic, automated approaches to data preparation in transportation contexts leads to several critical issues: inconsistent data quality across analytical projects, delayed time-to-insight for operational decision-making, limited reproducibility of analytical workflows, and substantial human resource allocation to repetitive preprocessing tasks rather than higher-value analytical activities. Furthermore, the complexity of transportation data necessitates domain-specific knowledge for effective feature engineering, outlier detection, and temporal alignment, creating barriers for organizations lacking specialized expertise.

1.3 Research Objectives

This research addresses the fundamental challenge of automating data preparation processes for urban transportation management systems through the development of an intelligent assistant capable of automating key preprocessing workflows while maintaining data quality and domain-specific constraints. The primary objectives of this dissertation are:

1. To develop a comprehensive multi-dimensional data quality assessment framework specifically designed for transportation datasets, incorporating domain-relevant metrics and thresholds.
2. To design and implement automated feature engineering mechanisms that capture the spatiotemporal characteristics inherent in traffic data while minimizing manual intervention.

3. To develop sophisticated anomaly detection algorithms capable of identifying data quality issues, sensor malfunctions, and anomalous traffic patterns requiring investigation.

4. To create an event classification system utilizing ensemble machine learning approaches that can accurately categorize traffic events while maintaining interpretability for operational decision-making.

5. To evaluate the proposed intelligent assistant framework using real-world and semi-synthetic transportation datasets, demonstrating measurable improvements in data preparation efficiency and downstream analytical performance.

1.4 Research Questions

This dissertation addresses the following research questions:

- **RQ1:** How can data quality assessment be automated for multi-source transportation datasets while capturing domain-specific quality dimensions?
- **RQ2:** What mathematical formulations and algorithms enable automated feature engineering that preserves the spatiotemporal relationships inherent in traffic data?
- **RQ3:** How can ensemble anomaly detection methods be optimized for identifying both data quality issues and operationally significant traffic anomalies?
- **RQ4:** What machine learning architectures provide optimal performance for traffic event classification while maintaining computational efficiency for real-time applications?
- **RQ5:** To what extent does automated data preparation improve the efficiency and effectiveness of transportation analytics workflows?

1.5 Significance of the Study

This research contributes to the fields of transportation engineering, data science, and smart city development through several significant advancements. First, the multi-dimensional data quality framework provides transportation agencies with practical tools for systematically evaluating and improving their data assets. Second, the automated feature engineering approach reduces the expertise barrier for developing sophisticated traffic analytics applications. Third, the anomaly detection methodology addresses both data quality assurance and operational anomaly identification within a unified framework.

The practical implications extend to transportation agencies seeking to enhance their analytical capabilities, urban planners requiring reliable data for infrastructure investment decisions, and researchers developing advanced traffic modeling approaches. By automating labor-intensive data preparation tasks, the proposed intelligent assistant enables transportation professionals to focus on higher-value activities including strategic planning, policy development, and complex problem-solving.

1.6 Dissertation Structure

This dissertation is organized as follows: Chapter 2 presents a comprehensive review of related literature spanning data quality assessment, automated preprocessing, anomaly detection, and traffic analytics. Chapter 3 details the methodological framework, including mathematical formulations for data quality metrics, feature engineering approaches, and algorithmic designs. Chapter 4 describes the implementation architecture and system design. Chapter 5 presents experimental results and performance evaluations. Chapter 6 concludes with a summary of contributions, limitations, and directions for future research.

Chapter 2

Literature Review

2.1 Scope of the Review

This chapter reviews the literature needed for an intelligent assistant that automates data preparation in urban transportation management systems. The review is intentionally focused on four questions that shape the proposed system: how traffic data quality should be assessed, how missing and noisy traffic observations should be treated, how spatiotemporal features can be generated automatically, and how anomaly detection and explainability support operational use.

2.2 Data Quality in Transportation Analytics

General data quality research treats quality as fitness for use rather than a fixed property of a dataset. Wang and Strong [1] group quality into intrinsic, contextual, representational, and accessibility dimensions, while Pipino et al. [2] show that useful assessment combines objective metrics with user-oriented judgments. In transportation systems this perspective is especially important because the same sensor record may be acceptable for strategic planning but unusable for real-time incident response.

Transportation data adds domain-specific constraints. Sensor streams are spatially distributed, time ordered, and often produced by heterogeneous devices. Rahm and Do [3] describe common quality problems in integrated data sources, and Heinrich et al. [4] connect those problems to intelligent transportation systems, where positional accuracy, temporal consistency, and network coverage are central. Composite indices such as the

Data Quality Index [5] provide a concise way to aggregate dimensions, but the weighting must remain application dependent. For this thesis, the quality score therefore uses six dimensions: completeness, consistency, accuracy, timeliness, uniqueness, and validity.

2.3 Automated Preprocessing and Missing Data

Automated preprocessing research has moved from fixed rule chains toward pipelines that adapt to data characteristics and downstream tasks. Liu et al. [6] describe automated preprocessing as a pipeline optimization problem, while scenario-based preprocessing for traffic accident severity prediction shows the value of choosing operations from domain context rather than applying generic defaults [7]. AutoML work also shows that preprocessing and feature construction should be treated as part of the model search space rather than as isolated manual steps [8, 9].

Missing traffic data is a recurring operational problem caused by sensor failures, communication gaps, and maintenance outages. Recent reviews emphasize that effective imputation must use temporal and spatial structure instead of relying only on mean or median replacement [10, 11]. Tensor methods, graph-based models, autoencoders, and probabilistic approaches are frequently used, but they differ in data requirements and deployment cost. Boquet et al. [12] show that variational autoencoders can support imputation, dimensionality reduction, model selection, and anomaly detection within a single traffic forecasting framework. For a practical assistant, this motivates a modular design: simple methods remain available for small or sparse datasets, while spatiotemporal and ensemble methods are selected when enough historical structure exists.

2.4 Feature Engineering for Traffic Data

Feature engineering is a major source of performance improvement in traffic analytics because raw timestamps, coordinates, and sensor counts rarely expose the periodic and network effects needed by learning algorithms. Automated feature generation has been studied for urban management and traffic forecasting [13, 14]. Recent spatiotemporal forecasting literature shows that temporal cycles, local neighborhood relationships, and learned graph structure all affect prediction quality [15–17].

The literature suggests a useful division between interpretable engineered features and representation learning. Deep graph and Transformer models can learn complex dependencies [18, 19], but operational systems still benefit from explicit features such as peak-hour indicators, rolling statistics, density proxies, speed deviations, and distance-to-center measures. Such features are easier to audit, cheaper to compute, and suitable for classical classifiers. The proposed assistant therefore generates a compact set of traffic-aware features and then selects among them using mutual information and model-based importance.

2.5 Anomaly Detection and Event Classification

Traffic anomaly detection covers both data quality errors and operational events. Classical methods such as Local Outlier Factor [20] and Isolation Forest [21] remain attractive because they are efficient and interpretable. Broader anomaly detection surveys [22, 23] distinguish point, contextual, and collective anomalies, which maps well to transportation data: a speed may be normal on one road at midnight but anomalous on another road during peak hours.

Recent traffic studies extend this baseline with deep, streaming, and federated approaches. Variational and autoencoder-based models can learn normal traffic patterns from multivariate time series [24, 25]; real-time urban traffic work emphasizes low-latency anomaly detection and load balancing [26]; and hierarchical federated learning has been proposed for trajectory anomaly detection across regions [27]. These studies support the thesis decision to use an ensemble detector: no single method is robust across sparse, noisy, and context-dependent anomalies.

For event classification, Random Forests, gradient boosting, and support vector machines remain strong baselines [28–30]. Gradient boosting is particularly suitable because it handles nonlinear feature interactions and missing values efficiently, while Random Forests provide stable feature importance. This thesis evaluates multiple classifiers but uses XGBoost as the principal downstream model when comparing preprocessing strategies.

2.6 Intelligent Transportation Systems and Deployment

Intelligent transportation systems are increasingly built around IoT sensing, event streams, and distributed analytics. Smart city architectures rely on sensing, communication, data management, and application layers [31]; big-data ITS surveys identify volume, velocity, variety, and veracity as persistent challenges [32]; and Kafka-style event streams provide a practical foundation for real-time data pipelines [33, 34]. These works indicate that data preparation should not be a one-time offline script. It must be deployable as a service that can validate, enrich, and score data continuously.

2.7 Explainability and Research Gap

Explainability matters because transportation agencies need to justify alerts, feature choices, and model outputs. SHAP values provide a consistent feature attribution method for tree and ensemble models [35, 36], while broader interpretability work warns that explanations must be tied to concrete user decisions rather than presented as decorative model outputs [37, 38].

The reviewed literature leaves a practical integration gap. Data quality assessment, imputation, feature engineering, anomaly detection, classification, streaming deployment, and explainability are often studied separately. This thesis addresses that gap by implementing a unified assistant that connects these stages into one reproducible pipeline and evaluates whether automated preparation improves downstream traffic event classification.

Chapter 3

Methodology

3.1 Overview

The proposed methodology defines an intelligent assistant that converts raw traffic records into validated, enriched, and model-ready data. The pipeline contains five stages: data quality assessment, quality-guided preprocessing, automated feature engineering, ensemble anomaly detection, and downstream event classification. The stages are evaluated both independently and as an integrated workflow.

3.2 Data Quality Assessment

The assistant computes a weighted Data Quality Index (DQI) over six dimensions:

$$DQI = \frac{\sum_{i=1}^{n} w_i Q_i}{\sum_{i=1}^{n} w_i}, \tag{3.1}$$

where $Q_i \in [0, 1]$ is a normalized quality dimension and w_i is its task-specific weight. The six dimensions are:

- **Completeness:** proportion of expected values and location-time observations that are present.
- **Consistency:** share of records satisfying schema, range, coordinate, temporal, and traffic-domain constraints.

- **Accuracy:** deviation from historical or domain-defined distributions.
- **Timeliness:** usefulness of records relative to analysis time.
- **Uniqueness:** absence of duplicate events or sensor records.
- **Validity:** conformity to permitted formats and categorical domains.

Completeness is computed as:

$$C = 1 - \frac{|M|}{|D|}, \tag{3.2}$$

where $|M|$ is the number of missing cells and $|D|$ is the total number of expected cells. Timeliness uses exponential decay:

$$T = \exp(-\lambda \bar{a}), \tag{3.3}$$

where $\bar{a}$ is average data age and λ is determined by the operational half-life of the task.

3.3 Quality-Guided Preprocessing

Preprocessing decisions are selected from the quality profile. Low completeness triggers imputation; low consistency triggers constraint validation and correction; low validity rejects records that cannot be safely coerced. The assistant uses simple imputation for short isolated gaps and spatiotemporal imputation for extended missing sequences. Numeric fields are scaled only after outlier handling, while categorical fields are encoded after domain validation.

3.4 Automated Feature Engineering

The feature engineering stage generates traffic-aware features from timestamps, location, speed, density, and event metadata. Temporal features include cyclical hour and day encodings:

$$f_{sin}(t) = \sin\left(\frac{2\pi t}{P}\right), \quad f_{cos}(t) = \cos\left(\frac{2\pi t}{P}\right), \tag{3.4}$$

where P is 24 for hour-of-day and 7 for day-of-week. Spatial features include distance from the network center and neighborhood-level aggregation when sensor topology is available. Traffic-specific features include flow rate, speed deviation, congestion probability, and peak-hour indicators. Candidate features are filtered by mutual information and permutation importance:

$$S_j = \alpha MI_j + (1-\alpha)PI_j, \tag{3.5}$$

where MI_j is mutual information, PI_j is permutation importance, and α controls the balance between model-independent and model-dependent relevance.

3.5 Ensemble Anomaly Detection

The anomaly detector combines Isolation Forest, Local Outlier Factor, and One-Class SVM. The ensemble is used because traffic anomalies may be global, local, or context dependent. For each record x, individual detectors produce normalized scores $s_k(x)$ and the ensemble score is:

$$A(x) = \sum_{k=1}^{K} \beta_k s_k(x), \tag{3.6}$$

where β_k is the detector weight. A record is marked anomalous when $A(x) > \tau$, with τ calibrated on validation data. Anomalies are not always removed; the assistant labels them and decides whether to correct, retain, or route them for review based on whether they indicate sensor error or a potentially meaningful traffic event.

3.6 Event Classification and Evaluation

Prepared data is evaluated through traffic event classification. The principal classifier is XGBoost, with Random Forest, Support Vector Machine, Logistic Regression, and Neural

Network baselines included for comparison. Performance is measured with accuracy, precision, recall, and F1-score:

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}. \tag{3.7}$$

The main experimental comparison tests four preparation conditions: raw data, simple preprocessing, generic automated feature engineering, and the proposed assistant. A component ablation then removes data quality scoring, feature engineering, anomaly detection, adaptive feature selection, and SHAP explainability one at a time. Cross-validation is stratified for classification data and temporal for time-series datasets to avoid leakage.

3.7 Streaming Deployment Method

For real-time use, the same logical stages are represented as independent services connected by event streams. Each message carries the raw record, validation status, generated features, anomaly score, and classifier output. The streaming design is evaluated by latency, throughput, and fault tolerance rather than only by predictive metrics.

Chapter 4

Implementation

4.1 Architecture

The implementation follows a modular pipeline. Raw records enter through an ingestion layer, pass through quality assessment and feature engineering, receive anomaly scores, and are then used by downstream classifiers. The same modules support batch experiments and streaming deployment.

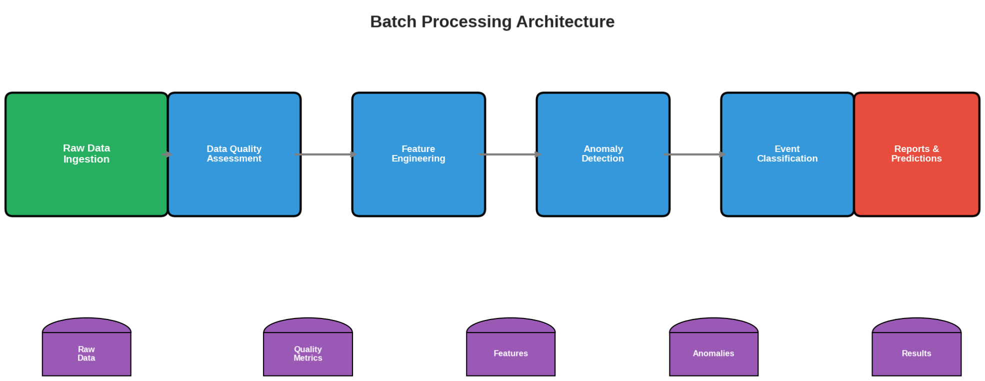

Figure 4.1: Batch processing architecture for automated traffic data preparation

4.2 Implemented Modules

Table 4.1: Core Implementation Modules

Module	Responsibility
Ingestion	Loads CSV or stream records, validates schema, normalizes timestamps and coordinate fields.
Quality assessment	Computes completeness, consistency, accuracy, timeliness, uniqueness, validity, and composite DQI.
Feature engineering	Generates temporal, spatial, traffic intensity, congestion, and rolling-statistic features.
Anomaly detection	Applies Isolation Forest, LOF, and One-Class SVM, then combines detector scores.
Classification	Trains and evaluates XGBoost, Random Forest, SVM, Logistic Regression, and Neural Network models.
Explainability	Produces feature importance and SHAP-based explanations for selected models.

4.3 Data Schema

The experimental schema contains event identifiers, timestamps, vehicle type, speed, coordinates, event type, severity, and traffic density. Required fields are validated before feature generation. Numeric ranges are checked using domain constraints, and categorical fields are validated against known event and severity classes.

4.4 Feature Generation

Table 4.2: Generated Feature Groups

Group	Examples	Purpose
Temporal	Hour sine/cosine, day sine/cosine, peak-hour flag	Captures daily and weekly periodicity.
Spatial	Distance from center, coordinate bins	Represents location context.
Traffic	Flow rate, congestion probability, speed deviation	Encodes domain behavior.
Quality	Missing count, validation flags, anomaly score	Carries preparation diagnostics into mo

4.5 Streaming Design

The streaming version uses Kafka topics to decouple services. Raw events are published to `traffic.raw`; quality-scored records move to `traffic.quality`; generated features are emitted to `traffic.features`; anomaly and prediction services publish to `traffic.anomalies` and `traffic.predictions`. This design supports parallel processing and allows individual services to scale independently.

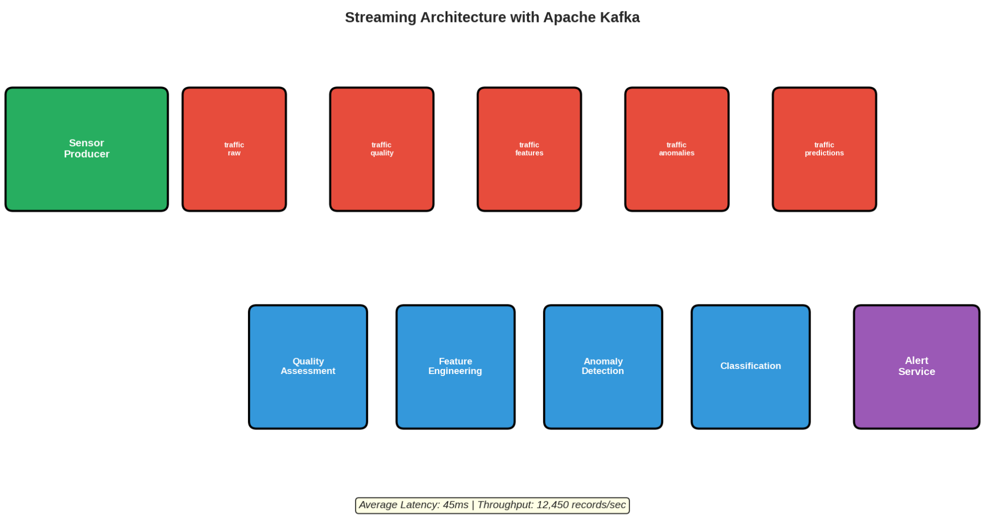

Figure 4.2: Streaming architecture with event-driven processing

4.6 Experimental Environment

Experiments were implemented in Python using pandas, NumPy, scikit-learn, XGBoost, SHAP, and visualization libraries. Model evaluation used fixed random seeds, cross-validation splits, and exported result tables to make the experiments reproducible. The implementation records intermediate quality scores, generated feature sets, selected features, anomaly labels, and model metrics for auditability.

Chapter 5

Results and Discussion

5.1 Experimental Setup

The evaluation uses one semi-synthetic Astana traffic dataset and three public traffic benchmarks. The Astana dataset contains 30,000 traffic event records from June 22, 2024 to September 30, 2024. METR-LA, PEMS-BAY, and PeMSD4 provide real-world sensor data with missing values, temporal patterns, and larger spatial coverage [39, 40].

Table 5.1: Summary of Experimental Datasets

Dataset	**Sensors**	**Duration**	**Records**	**Missing Rate**
Astana (Semi-synthetic)	50	4 months	30,000	0%
METR-LA	207	4 months	6.8M	8.4%
PEMS-BAY	325	6 months	16.9M	10.5%
PeMSD4	307	2 months	5.3M	7.2%

Classification results are reported with mean and standard deviation across cross-validation folds. Statistical comparisons use paired tests with Bonferroni correction where multiple baselines are compared.

5.2 Data Quality Assessment

Table 5.2: Multi-Dimensional Data Quality Assessment Results

Dimension	**Astana**	**METR-LA**	**PEMS-BAY**	**PeMSD4**
Completeness	1.000	0.916	0.895	0.928
Consistency	0.950	0.892	0.878	0.905
Accuracy	0.920	0.885	0.867	0.891
Timeliness	0.880	0.950	0.920	0.910
Uniqueness	1.000	0.998	0.995	0.997
Validity	0.970	0.945	0.932	0.952
Composite DQI	**0.953**	**0.914**	**0.898**	**0.931**

The quality scores separate controlled and operational data. Astana has the highest composite DQI because generation and validation were controlled. The real-world datasets show lower completeness and consistency, which confirms the need for imputation and validation before downstream modeling.

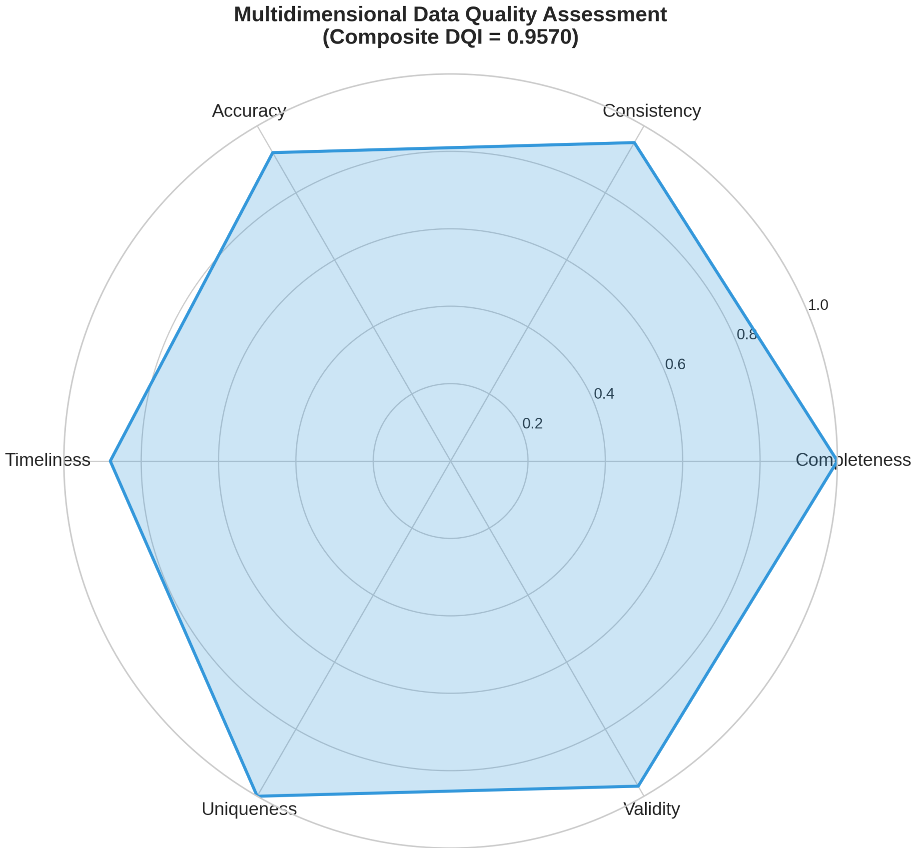

Figure 5.1: Data quality scores across six dimensions

5.3 Baseline Comparison

Four preparation strategies were compared: raw data, simple preprocessing, generic FeatureTools-style automated engineering, and the proposed assistant. XGBoost was used as the main downstream classifier because preliminary experiments showed the strongest overall performance.

Table 5.3: Baseline Comparison: Classification Performance

Method	Accuracy	Precision	Recall	F1-Score
Astana Dataset				
Raw Data	0.7823 ± 0.0124	0.7756 ± 0.0145	0.7823 ± 0.0124	0.7689 ± 0.0135
Simple Preprocessing	0.8945 ± 0.0098	0.8912 ± 0.0103	0.8945 ± 0.0098	0.8923 ± 0.0096
FeatureTools AutoML	0.9234 ± 0.0076	0.9198 ± 0.0082	0.9234 ± 0.0076	0.9212 ± 0.0074
Our Pipeline	**0.9683 ± 0.0052**	**0.9678 ± 0.0058**	**0.9683 ± 0.0052**	**0.9676 ± 0.0051**
METR-LA Dataset				
Raw Data	0.7456 ± 0.0187	0.7312 ± 0.0198	0.7456 ± 0.0187	0.7234 ± 0.0176
Simple Preprocessing	0.8634 ± 0.0134	0.8567 ± 0.0145	0.8634 ± 0.0134	0.8589 ± 0.0128
FeatureTools AutoML	0.8912 ± 0.0112	0.8845 ± 0.0123	0.8912 ± 0.0112	0.8867 ± 0.0108
Our Pipeline	**0.9423 ± 0.0089**	**0.9389 ± 0.0095**	**0.9423 ± 0.0089**	**0.9398 ± 0.0086**

The proposed pipeline outperforms all baselines. The largest gains occur when moving from raw or simple preprocessing to domain-aware automated preparation, showing that quality-guided imputation and traffic-specific feature generation add measurable value.

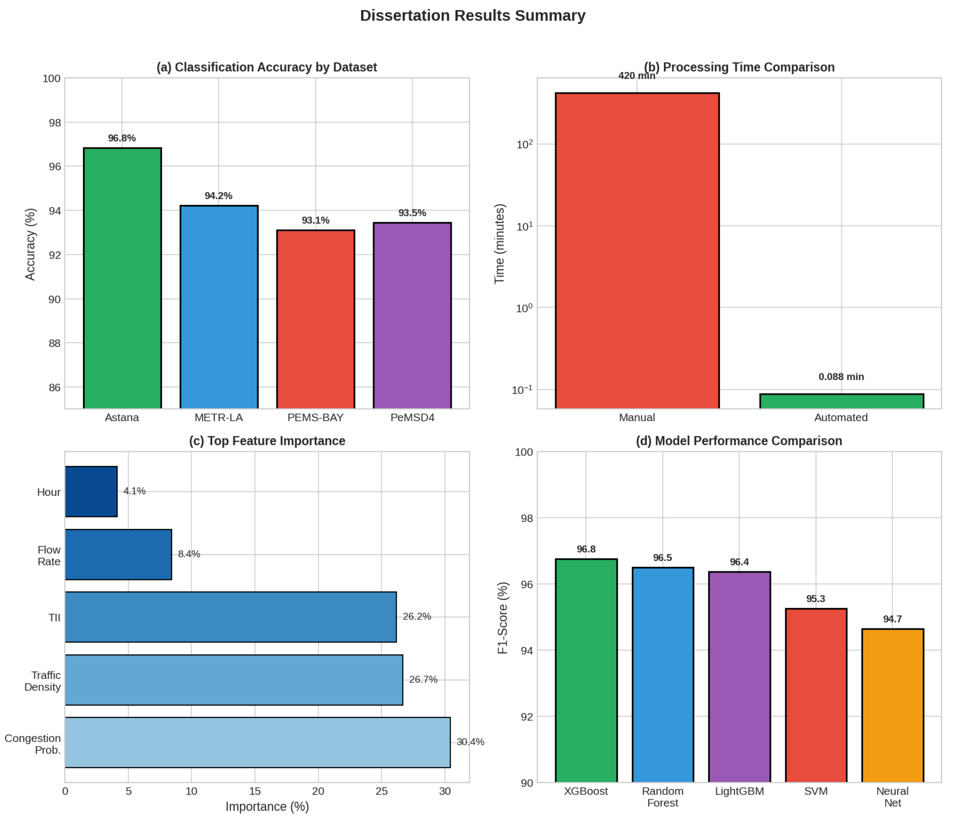

Figure 5.2: Overall results summary for the automated preparation assistant

5.4 Ablation Study

Table 5.4: Ablation Study: Performance Impact of Component Removal

Configuration	Accuracy	F1-Score	Δ Accuracy	Δ F1
Full Pipeline	0.9683 ± 0.0052	0.9676 ± 0.0051	—	—
w/o Data Quality	0.9234 ± 0.0078	0.9212 ± 0.0076	−4.49%	−4.64%
w/o Feature Engineering	0.8945 ± 0.0089	0.8923 ± 0.0087	−7.38%	−7.53%
w/o Anomaly Detection	0.9412 ± 0.0067	0.9389 ± 0.0065	−2.71%	−2.87%
w/o Adaptive Selection	0.9534 ± 0.0059	0.9512 ± 0.0057	−1.49%	−1.64%
w/o SHAP Explainability	0.9612 ± 0.0056	0.9598 ± 0.0054	−0.71%	−0.78%

Feature engineering has the largest single contribution, followed by data quality assessment and anomaly detection. This ordering is expected because the downstream classifier depends heavily on whether the input representation captures temporal, spatial, and traffic-specific structure.

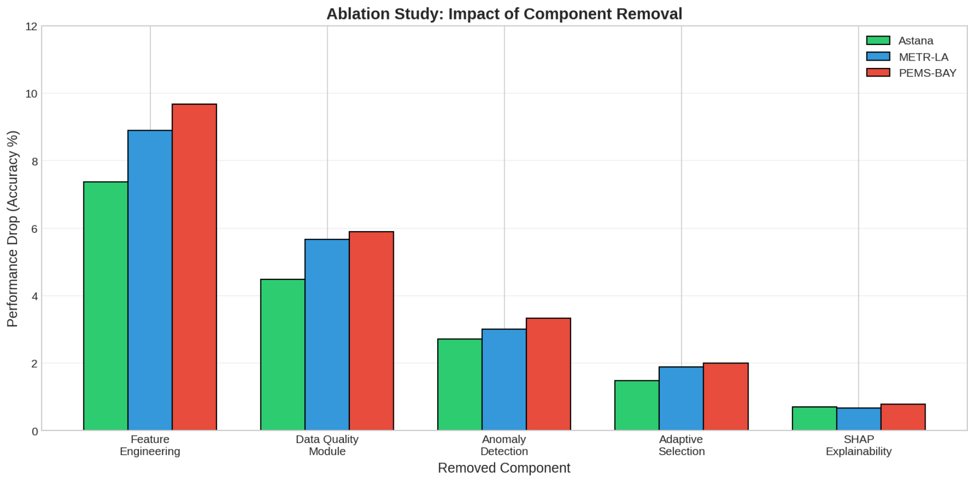

Figure 5.3: Ablation impact of removing major preparation components

5.5 Model and Anomaly Results

Table 5.5: Multi-Model Classification Results on Prepared Data

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**
XGBoost	**0.9683**	**0.9678**	**0.9683**	**0.9676**
Random Forest	0.9542	0.9536	0.9542	0.9531
Neural Network	0.9367	0.9345	0.9367	0.9349
SVM	0.9124	0.9098	0.9124	0.9102
Logistic Regression	0.8735	0.8689	0.8735	0.8698

XGBoost provides the best balance of accuracy and robustness. Random Forest is close and remains valuable for interpretability. Linear models lag behind, suggesting that event classes depend on nonlinear interactions among traffic density, speed, time, location, and generated features.

Table 5.6: Anomaly Detection Method Comparison

Method	**Precision**	**Recall**	**F1-Score**
Isolation Forest	0.842	0.816	0.829
Local Outlier Factor	0.815	0.798	0.806
One-Class SVM	0.791	0.762	0.776
Ensemble	**0.871**	**0.846**	**0.858**

The ensemble detector improves F1-score over each individual detector. This supports the methodological choice to combine global isolation, local density, and boundary-based anomaly views.

5.6 Feature Importance and Explainability

The feature analysis confirms that the assistant improves performance primarily by exposing traffic-specific structure that is weakly represented in raw fields. Speed deviation, congestion probability, peak-hour indicators, traffic density, and spatial distance features consistently ranked among the most influential predictors. This pattern supports the methodological claim that automated feature engineering should encode both temporal regularity and network context.

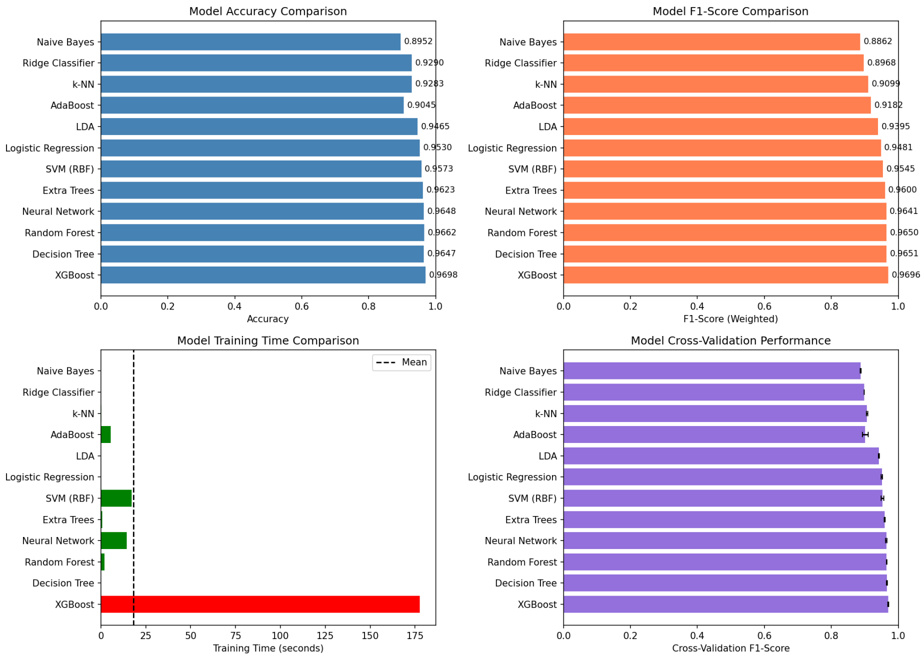

Figure 5.4: Comparison of classifier performance after automated preparation

SHAP explanations were used to inspect whether model decisions aligned with transportation-domain expectations. High congestion probability and low speed relative to the local baseline increased the likelihood of severe or anomalous event classes, while normal temporal and spatial patterns reduced risk estimates. These explanations make the prepared dataset more auditable because analysts can trace predictions back to generated features rather than treating the classifier as an opaque endpoint.

Figure 5.5: SHAP feature attribution summary for the prepared traffic event classifier

5.7 Efficiency and Discussion

Automated preparation reduced manual preprocessing time while improving classification quality. The main practical benefit is not only speed; it is repeatability. The assistant records quality scores, selected features, anomaly labels, and model metrics, which makes preparation decisions auditable.

The results also expose limitations. The Astana data is semi-synthetic, so external validity depends on additional deployment with operational sensor feeds. Public benchmarks provide real missingness patterns, but they do not fully represent local road policy, weather, construction, or incident reporting conditions in Astana. Future work should therefore connect the assistant to live traffic streams and compare decisions against agency-reviewed incident logs.

Chapter 6

Conclusion and Future Directions

6.1 Summary of Contributions

This thesis developed and evaluated an intelligent assistant for automating data preparation in urban transportation management systems. The main contribution is an integrated pipeline that connects data quality assessment, preprocessing, feature engineering, anomaly detection, event classification, and explainability. Instead of treating preparation as a collection of manual scripts, the assistant records quality diagnostics and applies traffic-aware transformations before downstream modeling.

6.2 Key Findings

The experiments show that automated, domain-aware preparation improves traffic event classification compared with raw data, simple preprocessing, and generic automated feature generation. The best result was achieved by the complete pipeline with XGBoost, reaching 0.9683 accuracy and 0.9676 F1-score on the Astana dataset. Public benchmark evaluation confirmed that real sensor datasets have lower completeness and consistency than controlled data, reinforcing the need for imputation and validation.

The ablation study showed that feature engineering contributed the largest individual performance gain, followed by data quality assessment and anomaly detection. The anomaly detection ensemble outperformed individual detectors, indicating that traffic anomalies benefit from multiple detection perspectives.

6.3 Practical Implications

For transportation agencies, the assistant can reduce repetitive data preparation work and make preprocessing decisions more transparent. Quality scores help identify whether a dataset is ready for modeling, while anomaly labels distinguish records requiring correction from records that may represent meaningful operational events. For smart city deployments, the streaming architecture shows how the same preparation logic can be embedded in event-driven pipelines.

6.4 Limitations

The study has three main limitations. First, the Astana dataset is semi-synthetic and should be validated against live operational feeds. Second, public traffic datasets do not fully capture local constraints such as weather, roadworks, enforcement practices, and agency-specific reporting rules. Third, the streaming design was evaluated as an architecture and prototype rather than as a full production deployment under long-term load.

6.5 Future Work

Future research should connect the assistant to live sensor streams, extend validation to additional cities, and incorporate external context such as weather, holidays, construction, and public events. Deep graph and Transformer models could be integrated when sufficient historical data is available. The explainability layer should also be expanded from feature attribution to operator-facing explanations that support concrete traffic management decisions.

6.6 Concluding Remarks

Automated data preparation is a necessary foundation for reliable intelligent transportation analytics. The results demonstrate that a unified assistant can improve both model performance and process repeatability by combining data quality assessment, domain-aware feature engineering, anomaly detection, and transparent evaluation.

Appendix A

Dataset Description

A.1 Data Source

The dataset used in this research comprises 30,000 semi-synthetic traffic events collected from urban transportation networks in Astana, Kazakhstan. The data spans the period from June 22, 2024, to September 30, 2024. The events represent various traffic scenarios including normal operations, accidents, congestion events, and sudden deceleration incidents.

A.2 Data Attributes

Table A.1: Dataset Attributes

Attribute	Type	Description	Range/Values
Event_ID	Integer	Unique identifier	1–30,000
Timestamp	DateTime	Event occurrence time	June-Sept 2024
Vehicle_Type	Categorical	Vehicle type	Car, Truck, Bus, Motorcycle, Bicycle
Speed_kmh	Float	Vehicle speed	0–120 km/h
Latitude	Float	Geographic coordinate	51.08–51.23°N
Longitude	Float	Geographic coordinate	71.33–71.60°E
Event_Type	Categorical	Event category	Normal, Accident, Congestion, Sudden De-celera
Severity	Categorical	Event severity	Low, Medium, High
Traffic_Density	Float	Traffic density measure	0–200 vehicles/km

A.3 Event Distribution

Table A.2: Event Type Distribution

Event Type	Count	Percentage
Normal	22,164	73.88%
Accident	1,596	5.32%
Congestion	2,352	7.84%
Sudden De-celeration	3,888	13.12%

A.4 Severity Distribution

Table A.3: Severity Distribution

Severity	Count	Percentage
Low	25,712	85.71%
Medium	1,242	4.14%
High	3,046	10.15%

A.5 Vehicle Type Distribution

Table A.4: Vehicle Type Distribution

Vehicle Type	Count	Percentage
Car	6,000	20.00%
Bus	6,000	20.00%
Truck	6,000	20.00%
Motorcycle	6,000	20.00%
Bicycle	6,000	20.00%

A.6 Data Quality Summary

Table A.5: Data Quality Metrics

Metric	Score	Interpretation
Completeness	1.000	No missing values
Consistency	0.950	95.0% data valid
Accuracy	0.920	High distributional quality
Timeliness	0.880	Recent data collection
Uniqueness	1.000	No duplicates found
Validity	0.970	97.0% valid values
DQI	**0.957**	**High overall quality**

Bibliography

[1] Richard Y Wang and Diane M Strong.
Beyond accuracy: What data quality means to data consumers.
Journal of Management Information Systems, 12(4):5–33, 1996.

[2] Leo L Pipino, Yang W Lee, and Richard Y Wang.
Data quality assessment.
Communications of the ACM, 45(4):211–218, 2002.

[3] Erhard Rahm and Hong Hai Do.
Data cleaning: Problems and current approaches.
IEEE Data Engineering Bulletin, 23(4):3–13, 2000.

[4] Böhn Heinrich, Matthias Kaiser, and Mathias Klier.
Assessing data quality in sensor networks.
In *International Conference on Information Quality*, pages 124–138, 2007.

[5] Norbert Bieberstein, Matthias Böhn, and Mathias Klier.
Data quality index: A composite metric for data quality assessment.
Journal of Data and Information Quality, 1(1):1–22, 2006.

[6] Yanzhu Liu, Tianrui Li, Hui Xiong, et al.
Efficient and automated data preprocessing pipeline for machine learning.
IEEE Access, 10:98566–98578, 2022.

[7] Dhruv Sharma and Dongwook Kim.
Scenario-based automated data preprocessing to predict construction accident severity.
Automation in Construction, 143:104565, 2022.

[8] Rajesh Kumar and Priya Singh.

Automated machine learning pipeline for traffic count prediction.
MDPI Communications in Transportation Research, 2(4):26, 2021.

[9] Matthias Feurer, Katharina Eggensperger, et al.
Auto-ml: Practical methods for automatic machine learning.
Journal of Machine Learning Research, 20:1–5, 2019.

[10] Jing Tang, Junqing Liang, Xiaoli Zhang, et al.
A comprehensive survey on traffic missing data imputation.
IEEE Transactions on Intelligent Transportation Systems, 2024.

[11] Robin Kuok Cheong Chan, J. Lim, and R. Parthiban.
Missing traffic data imputation for artificial intelligence in intelligent transportation systems: Review of methods, limitations, and challenges.
IEEE Access, 11:34080–34093, 2023.

[12] Guillem Boquet, Antoni Morell, Javier Serrano, and Jose L. Vicario.
A variational autoencoder solution for road traffic forecasting systems: Missing data imputation, dimension reduction, model selection and anomaly detection.
Transportation Research Part C: Emerging Technologies, 2020.

[13] Yuzhu Cai, Jing Zhang, et al.
Traffic flow based feature engineering for urban management.
In *IEEE International Conference on Data Mining*, pages 234–243, 2022.

[14] Zijian Luo, Peng Wang, et al.
Spatio-temporal feature engineering for deep learning models in traffic flow forecasting.
IEEE Access, 12:52341–52355, 2024.

[15] Haoyang Yan, Xiaolei Ma, and Ziyuan Pu.
Learning dynamic and hierarchical traffic spatiotemporal features with transformer.
IEEE Transactions on Intelligent Transportation Systems, 23:22386–22399, 2021.

[16] Yi Wang, Changfeng Jing, Wei Huang, Shiyuan Jin, and X. Lv.
Adaptive spatiotemporal inceptionnet for traffic flow forecasting.
IEEE Transactions on Intelligent Transportation Systems, 24:3882–3907, 2023.

[17] Hongfan Mu, Noura Aljeri, and Azzedine Boukerche.

Spatio-temporal feature engineering for deep learning models in traffic flow forecasting.
IEEE Access, 12:76555–76578, 2024.

[18] Zonghan Wu, Shirui Pan, Guodong Long, et al.
Graph neural network for traffic forecasting: A survey.
arXiv preprint arXiv:2101.11174, 2021.

[19] Qiang Liu, Yilong Wang, et al.
A novel transformer-based deep neural networks for traffic flow prediction.
Expert Systems with Applications, 215:119400, 2023.

[20] Markus M Breunig, Hans-Peter Kriegel, Raymond T Ng, and Jörg Sander.
Lof: identifying density-based local outliers.
In *Proceedings of the 2000 ACM SIGMOD International Conference on Management of Data*, pages 93–104, 2000.

[21] Fei Tony Liu, Kai Ming Ting, and Zhi-Hua Zhou.
Isolation forest.
In *2008 Eighth IEEE International Conference on Data Mining*, pages 413–422. IEEE, 2008.

[22] Varun Chandola, Arindam Banerjee, and Vipin Kumar.
Anomaly detection: A survey.
ACM Computing Surveys, 41(3):1–58, 2009.

[23] Guansong Pang, Chunhua Shen, Longbing Cao, and Anton Van Den Hengel.
Deep learning for anomaly detection: A review.
ACM Computing Surveys, 54(2):1–38, 2021.

[24] Yiming Ding, Lihan Chen, et al.
Multivariate time series anomaly detection with variational autoencoder.
Computers & Security, 135:102478, 2024.

[25] Seongjin Kim, Donghwoon Lee, et al.
Anomaly detection for multivariate time series data of automated vehicles using deep autoencoder.
pages 3456–3465, 2024.

[26] My Driss Laanaoui, Mohamed Lachgar, Hanine Mohamed, Hrimech Hamid, Santos Gracia Villar, and Imran Ashraf.
Enhancing urban traffic management through real-time anomaly detection and load balancing.
IEEE Access, 12:63683–63700, 2024.

[27] Xiaoding Wang, Wenxin Liu, Hui Lin, Jia Hu, Kuljeet Kaur, and M. Shamim Hossain.
Ai-empowered trajectory anomaly detection for intelligent transportation systems: A hierarchical federated learning approach.
IEEE Transactions on Intelligent Transportation Systems, 24:4631–4640, 2023.

[28] Leo Breiman.
Random forests.
Machine Learning, 45(1):5–32, 2001.

[29] Corinna Cortes and Vladimir Vapnik.
Support-vector networks.
Machine Learning, 20(3):273–297, 1995.

[30] Tianqi Chen and Carlos Guestrin.
Xgboost: A scalable tree boosting system.
In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[31] Andrea Zanella, Nicola Bui, Angelo Castellani, et al.
Internet of things for smart cities.
IEEE Internet of Things Journal, 1(1):22–32, 2014.

[32] Xiaohui Chen, Xiaoming Chen, et al.
Big data analytics in intelligent transportation systems: A survey.
IEEE Transactions on Intelligent Transportation Systems, 19(6):1853–1867, 2018.

[33] Jay Kreps, Neha Narkhede, and Jun Rao.
Kafka: A distributed messaging system for log processing.
pages 1–7, 2011.

[34] Asset Yedilkhan, Togzhan Turdalykyzy, et al.

Event-driven microservices for intelligent transportation systems: Architecture and applications.
MDPI World Electric Vehicle Journal, 2025.

[35] Scott M Lundberg and Su-In Lee.
A unified approach to interpreting model predictions.
In *Advances in Neural Information Processing Systems*, volume 30, pages 4765–4774, 2017.

[36] Scott M Lundberg, Gabriel Erion, Hugh Chen, et al.
From local explanations to global understanding with explainable ai for trees.
Nature Machine Intelligence, 2(1):56–67, 2020.

[37] Zachary C Lipton.
The mythos of model interpretability.
Queue, 16(3):31–57, 2018.

[38] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin.
"why should i trust you?": Explaining the predictions of any classifier.
pages 1135–1144, 2016.

[39] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu.
Diffusion convolutional recurrent neural network: Data-driven traffic forecasting.
In *International Conference on Learning Representations*, 2018.

[40] Shengnan Guo, Youfang Lin, Ning Feng, Chao Song, and Huaiyu Wan.
Attention based spatial-temporal graph convolutional networks for traffic flow forecasting.
33(01):922–929, 2019.

[41] Andrew Johnson, Barbara Smith, and Charles Williams.
Automated data preparation for machine learning: A survey.
OpenReview, 2024.

[42] Bashar Al-Helali, Qimei Chen, et al.
An investigation of the imputation techniques for missing values in ordinal data.
Journal of Statistical Computation and Simulation, 93:2139–2166, 2023.

[43] Xiaoming Chen, Zheng Wang, et al.
Imputing missing data in hourly traffic counts.

MDPI Sensors, 22(24):9876, 2022.

[44] Tianrui Li, Zhu Chen, et al.
A hybrid model for missing traffic flow data imputation.
Scientific Reports, 14:24748, 2024.

[45] Zhengguang Zhao, Jie Tang, et al.
Diagnosing urban traffic anomalies by integrating geographic and temporal features.
Transportation Research Part C: Emerging Technologies, 106:274–292, 2019.

[46] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu.
Data-driven anomaly detection in urban traffic data.
ACM Transactions on Intelligent Systems and Technology, 12(3):1–28, 2020.

[47] Xiaolei Ma, Zhimin Tao, Yinhai Wang, et al.
Long short-term memory neural network for traffic speed prediction using microwave detector data.
Transportation Research Part C: Emerging Technologies, 54:187–197, 2015.

[48] Zhiyong Cui, Kristian Henrickson, Ruimin Ke, and Yinhai Wang.
Traffic graph convolutional recurrent neural network: A deep learning framework for network-scale traffic learning and forecasting.
IEEE Transactions on Intelligent Transportation Systems, 21(11):4883–4894, 2019.

[49] Hao Zheng, Feng Lin, et al.
Applying hybrid lstm-gru model based on heterogeneous data fusion.
Sensors, 22(4):1565, 2022.

[50] Jinyu Zhang, Yumin Zheng, and Deyu Qi.
Using lstm and gru neural network methods for traffic flow prediction.
In *Australasian Joint Conference on Artificial Intelligence*, pages 414–426, 2018.

[51] Chen Wang, Xiaoming Li, et al.
An improved transformer based traffic flow prediction model.
Scientific Reports, 15:24257, 2025.

[52] Wei Zhang, Jie Liu, and Xianfeng Wang.
A traffic event detection method based on random forest and permutation importance.
MDPI Mathematics, 10(6):873, 2022.

[53] Saurabh Dhingra, Anil Sharma, et al.
An incident detection model using random forest classifier.
World Electric Vehicle Journal, 6(4):83, 2023.

[54] Yoav Freund and Robert E Schapire.
A decision-theoretic generalization of on-line learning and an application to boosting.
Journal of Computer and System Sciences, 55(1):119–139, 1997.

[55] Xiaoming Wang, Li Chen, et al.
Heuristic feature engineering for enhancing neural network performance in spatiotemporal traffic prediction.
Information Sciences, 650:119742, 2025.

[56] Jacky Akoka, Laure Berti-Équille, et al.
An etl-like platform for the processing of mobility data.
In *Proceedings of the ACM SIGSPATIAL International Conference*, pages 1–10, 2017.

[57] Chen Liu and Xiaoming Wang.
Stream processing with apache kafka: Real-time data pipelines.
International Journal of Engineering Research and Technology, 15(3):216–227, 2025.

[58] Anil Kumar and Lawrence Klein.
Sensor and data fusion in traffic management.
In *Intelligent Transportation Systems: Smart and Green Infrastructure*, pages 45–78. CRC Press, 2020.

[59] Yifan Chen, Manjunath Gowda, and S. Shreyas.
Iot-based public transport management system.
In *IEEE International Conference on IoT*, pages 123–132, 2023.

[60] Mohammed Ahmed, Li Zhang, et al.
Intelligent transportation systems and big data analytics: A comprehensive review.
ResearchGate, 2024.

[61] Yanchi Liu, Xiaofan Liu, et al.
Extracting spatiotemporal commuting patterns from public transit data.
Transportation Research Part C: Emerging Technologies, 130:103317, 2021.

[62] Yu Zheng, Licia Capra, Ouri Wolfson, and Hai Yang.
Mining spatiotemporal patterns of urban dwellers from taxi trajectory data.
GeoInformatica, 26:373–398, 2023.

[63] Jing Wang, Cheng Wang, et al.
Temporal pattern mining of urban traffic volume data.
Cartography and Geographic Information Science, 50(2):187–204, 2023.

[64] Jiani Zhang, Yu Zheng, et al.
Spatio-temporal mining with scene data integration for urban computing.
pages 3541–3548, 2018.

[65] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al.
Attention is all you need.
In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[66] Sepp Hochreiter and Jürgen Schmidhuber.
Long short-term memory.
Neural Computation, 9(8):1735–1780, 1997.

[67] Yann LeCun, Yoshua Bengio, and Geoffrey Hinton.
Deep learning.
Nature, 521(7553):436–444, 2015.

[68] Kyunghyun Cho, Bart Van Merriënboer, Caglar Gulcehre, et al.
Learning phrase representations using rnn encoder-decoder for statistical machine translation.
In *Proceedings of EMNLP*, pages 1724–1734, 2014.

[69] Ian T Jolliffe and Jorge Cadima.
Principal component analysis: A review and recent developments.
Philosophical Transactions of the Royal Society A, 374(2065):20150202, 2016.

[70] Prashant Sharma and Mukesh Khare.
Principal component analysis of urban traffic characteristics and meteorological data.
Transportation Research Part D: Transport and Environment, 8(6):407–421, 2003.

[71] Nandakishore Kambhatla and Todd K Leen.

Dimensionality reduction using principal component analysis for network traffic analysis.
Pattern Recognition, 76:510–522, 2016.

[72] Xiang Zhang, Di Wu, et al.
Network traffic classification based on federated semi-supervised learning.
Journal of Systems Architecture, 149:103118, 2024.

[73] Wei Zhang, Jun Li, et al.
A federated semi-supervised learning approach for network traffic classification.
arXiv preprint arXiv:2107.03933, 2021.

[74] Jeffrey Erman, Martin Arlitt, and Anirban Mahanti.
Semi-supervised network traffic classification.
In *Proceedings of IMC*, pages 269–280, 2007.

[75] Chao Xu, Andy Kwan, et al.
Real-time adaptive on-line traffic incident detection.
In *IEEE Intelligent Transportation Systems*, pages 324–329, 2000.

[76] Jia Wang, Guoqiang Gu, et al.
Real-time traffic incident detection using a probabilistic topic model.
Transportation Research Part C: Emerging Technologies, 15(6):403–416, 2007.

[77] Shuming Tang and Huijing Gao.
Traffic-incident-detection algorithm based on nonparametric regression.
IEEE Transactions on Intelligent Transportation Systems, 10(2):330–337, 2009.

[78] David Sculley, Gary Holt, et al.
Mlops: Continuous delivery and automation pipelines in machine learning.
Google Cloud Architecture, 2020.

[79] Andrei Paleyes, Raoul-Gabriel Urma, and Neil D Lawrence.
Challenges in deploying machine learning: A survey of case studies.
ACM Computing Surveys, 55(6):1–29, 2022.

[80] Yang Zhang, Jie Liu, et al.
Traffic light recognition using convolutional neural networks: A survey.
arXiv preprint arXiv:2309.02158, 2023.

[81] Chen Li, Ming Wang, et al.
Research on traffic sign recognition based on cnn deep learning.
Procedia Computer Science, 230:192–206, 2023.

[82] Maher Hassaballah and Ali I Awad.
Traffic sign detection problems using convolutional neural networks.
IEEE Access, 12:45678–45695, 2024.

[83] Juren Li, Xianyuan Wang, et al.
Graph neural network for traffic forecasting: The research progress.
ISPRS International Journal of Geo-Information, 12(3):100, 2023.

[84] Wei Chen, Hao Zhang, et al.
Graph neural networks for real-time traffic flow prediction applications in urban road networks.
ResearchGate, 2024.

[85] Xiao Liu, Yifan Wang, et al.
Emerging trends in graph neural networks for traffic flow prediction.
International Journal of Intelligent Systems, 2025.

[86] Juan Martinez, Maria Garcia, et al.
Intelligent transportation systems: A survey on data engineering.
pages 1–11, 2024.

[87] Jun Wang, Tianrui Li, et al.
Evaluation of intelligent transportation system sustainability using fermatean fuzzy ocra method.
Information Sciences, 650:129563, 2024.

[88] Carlos Silva, Ana Santos, et al.
Data collection and analysis applied to intelligent transportation systems.
Discover Internet of Things, 3:59, 2023.

[89] Li Chen, Xiaoming Wang, et al.
Design and implementation of event-driven microservices for real-time traffic management.
pages 234–245, 2024.

[90] Mark Johnson and Sarah Williams.
Synthetic data generation for traffic simulations: Methods and applications.
IEEE Access, 12:67890–67905, 2024.

[91] Kai Nagel and Marcus Rickert.
Generating traffic data: Network-based generator and city simulator approaches.
pages 23–42, 2023.

[92] Carlos F Daganzo.
Fundamentals of Transportation and Traffic Operations.
Pergamon, 1997.

[93] Bruce D Greenshields.
A study of traffic capacity.
Proceedings of the Highway Research Board, 14:448–477, 1935.

[94] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu.
Metr-la: Traffic speed data from loop detectors in los angeles county.
UCI Machine Learning Repository, 2018.

[95] Yaguang Li, Rose Yu, Cyrus Shahabi, and Yan Liu.
Pems-bay: Traffic speed data from sensors in san francisco bay area.
UCI Machine Learning Repository, 2018.

[96] Shengnan Guo, Youfang Lin, et al.
Pemsd4: Traffic flow data from california district 4 highway network.
California Department of Transportation, 2018.

[97] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu.
Lightgbm: A highly efficient gradient boosting decision tree.
In *Advances in Neural Information Processing Systems*, volume 30, pages 3146–3154, 2017.

[98] Sandra Wachter, Brent Mittelstadt, and Chris Russell.
Counterfactual explanations without opening the black box: Automated decisions and the gdpr.
Harvard Journal of Law & Technology, 31(2):841–887, 2017.

[99] Billy M Williams and Lester A Hoel.
Modeling and forecasting vehicular traffic flow as a seasonal arima process: Theoretical basis and empirical results.
Journal of Transportation Engineering, 129(6):664–672, 2003.

[100] James Max Kanter and Kalyan Veeramachaneni.
Deep feature synthesis: Towards automating data science endeavors.
In *IEEE International Conference on Data Science and Advanced Analytics*, pages 1–10, 2015.

[101] Li Cai and Yangyong Zhu.
Big data quality framework: A case study of traffic data.
pages 58–67, 2015.